

MANFAAT ARTIFICIAL INTELLIGENCE DI KALANGAN MAHASISWA

MANFAAT A.I

Pemanfaatan yang terlihat antara lain seperti A.I sebagai asisten digital dalam menampilkan tutorial pembelajaran, sebagai sistem dalam mengevaluasi mahasiswa, sebagai sistem dalam layanan obrolan mahasiswa, serta contoh-contoh lainnya.

DAMPAK POSITIF

1. Bisa menyimpan data yang tidak terbatas
2. Memiliki ketepatan dan keakuratan dalam mengerjakan tugasnya
3. Bisa digunakan kapan saja dan dalam jangka waktu yang panjang
4. Mudah digandakan dan disebarluaskan
5. Lebih Murah
6. Bisa mengerjakan pekerjaan dengan lebih cepat dan lebih baik

DAMPAK NEGATIF

1. Pengangguran Meningkat
2. Penyalahgunaan
3. Kurangnya keamanan
4. Privasi yang minim

HARAPAN

semua model pembelajaran teknologi ini diterapkan dan dikembangkan untuk meniadakan batas dan pemerataan pendidikan bagi semua, pengajar menjadi lebih produktif, pengajar menjadi bersifat individual, materi pembelajaran dapat diberikan dan diakses baik secara terjadwal atau bebas flexibel kapan saja



00.09



1

1 perangkat

126,1

KB/dtk



4G

 **JASA**

INSTALL ULANG LAPTOP & PC

📍 PANGGILAN SAMARINDA



LAPTOP / PC LEMOT, LELET,
SERING ERROR ATAU TERKENA VIRUS?

Saya Solusinya!

SEMUA WINDOWS SAYA PUNYA

Windows
XP

Windows 7



Windows 8.1



Windows 10



Windows 11

- ✓ INSTALL ULANG WINDOWS
- ✓ INSTALL DRIVER
- ✓ BERSIHKAN VIRUS
- ✓ OPTIMASI PERFORMA

Laptop/PC jadi cepat & stabil lagi!



HARGA TERJANGKAU

Yang penting laptop/PC
Anda normal lagi.

**GAK PERLU MAHAL KOK,
AMAN AJA!**



PANGGILAN KE LOKASI ANDA
di Samarinda & sekitarnya



Saldi Ahmad

SARANGI INSTALL & SOFTWARE

(Sesuai Kesepakatan)

KEMARIN PUKUL 17.40

SAMBIL NGOBROL BISA,

mau tanya-tanya juga boleh!

**BIAR URUSAN KOMPUTER
JADI MUDAH!**

JASA INSTALL ULANG LAPTOP & PC –

PANGGILAN SAMARINDA

0823-5033-1742

(CALL • SMS • WHATSAPP)

Laptop atau PC mulai lemot, ser... Lihat lainnya



ner · Big Boom





⚡ **FORMULA FRIDAY** ⚡

VOLTAGE DROP FORMULA

Important for proper cable sizing and efficient power distribution.



FORMULA

$$V_d = \sqrt{3} \times I \times L \times R$$

Where,

V_d = Voltage Drop (Volt)

I = Current (Ampere)

L = Length of Cable (meter)

R = Resistance of Cable per meter (Ohm/meter)



Remember

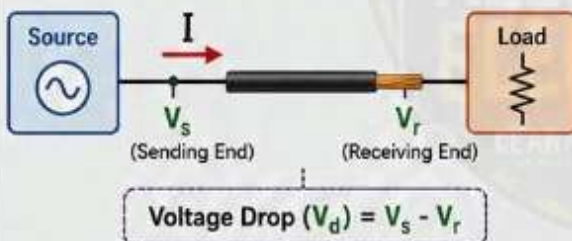
Lower the voltage drop, higher the efficiency and better the performance.

$\sqrt{3} = 1.732$

(For Three Phase System)

WHAT DOES IT MEAN?

Voltage drop is the loss of voltage in a cable due to its resistance when current flows through it.



EXAMPLE

A 3-phase motor draws 25 A current. The cable length is 50 meters (one way). The resistance of cable is 0.0004 Ω /m. Calculate the voltage drop.

Given:

$I = 25$ A

$L = 50$ m

$R = 0.0004$ Ω /m

Using Formula:

$$\begin{aligned} V_d &= \sqrt{3} \times I \times L \times R \\ &= 1.732 \times 25 \times 50 \times 0.0004 \\ &= 0.866 \text{ Volt} \end{aligned}$$

Voltage Drop = 0.87 V (Approx.)

HOW TO REDUCE VOLTAGE DROP?

- ✓ Use cable with larger cross-sectional area.
- ✓ Keep cable length as short as possible.
- ✓ Use low resistance cable (copper preferred).
- ✓ Balance the load in three phase system.



RECOMMENDED VOLTAGE DROP

- For Lighting Circuits : $\leq 3\%$
- For Power Circuits : $\leq 5\%$
- For Motors : $\leq 5\%$

WHERE IS IT USED?



To select the correct cable size for installations.



To ensure proper voltage is available at the load.



To improve system efficiency and performance.



To prevent overheating and energy loss.



SAFETY TIP

High voltage drop causes low performance, overheating, and extra energy loss.
Always calculate and maintain within limit.



Follow standards like IEC, IS & NEC for safe design.



QUESTION FOR YOU

What is the maximum recommended voltage drop for power circuits?

COMMENT BELOW ↓



SHARE THIS POST

Share with fellow electricians and help them learn and grow.

SHARE NOW ➔



FOLLOW US

Follow ALLROUNDER ELECTRICIAN for daily electrical knowledge and practical tips.

FOLLOW NOW 👤



LEARN • SHARE • GROW



What is KVAR ?

Date : ___/___/___

Page No. : _____

★ **KVAR** stands for Kilo Volt Ampere Reactive.

It is the unit of **Reactive Power** in an AC Electrical System.

★ Reactive power does not perform any useful work, but it is necessary to create magnetic field in inductive equipment like motors, transformers, reactors, etc.

★ Types of Power

Power	Symbol	Unit	Purpose
Active Power	P	kW	Performs Useful Work
Reactive Power	Q	kVAR	Creates Magnetic / Electric Fields
Apparent Power	S	kVA	Total Power Supplied

Unit

- Unit of Reactive Power = **VAR** (Volt Ampere Reactive)
- 1000 VAR = **1 KVAR**

Simple Formula

$$Q \text{ (kVAR)} = V \times I \times \sin \phi$$

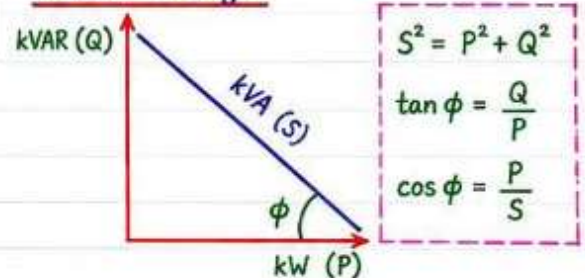
Where,

V = Voltage (Volt)

I = Current (Ampere)

ϕ = Phase Angle between Voltage & Current

Power Triangle



Where is kVAR Used ?

- Induction Motors 
- Power Transformers 
- Air Conditioners 
- Pumps 
- Fans 
- Welding Machines 
- Fluorescent Lighting with Chokes 

Why is kVAR Important ?

- ✓ Improves Power Factor when compensated.
- ✓ Reduces current in cables.
- ✓ Lowers transmission losses.
- ✓ Improves voltage regulation.
- ✓ Helps avoid utility Power Factor penalties.



Example

A motor consumes:

Active Power (P) = 100 kW

Power Factor ($\cos \phi$) = 0.8 Lagging

Then,

$$\text{Apparent Power, } S = \frac{P}{\cos \phi} = \frac{100}{0.8} = 125 \text{ kVA}$$

$$\text{Reactive Power, } Q = P \times \tan \phi$$

$$\phi = \cos^{-1} 0.8 = 36.87^\circ, \tan \phi = 0.75$$

$$Q = 100 \times 0.75 = 75 \text{ kVAR}$$



How to Reduce kVAR Demand ?

Install **Capacitor Bank**.

Capacitors supply reactive power locally, reduces the amount of kVAR drawn from the electrical grid and improves the power factor.



Easy to Remember

kW = Useful Power
(Does the Work)

kVAR = Magnetic Power
(Supports the Work)

kVA = Total Power
Supplied

★ **Note** : High kVAR means low Power Factor.
Always try to keep Power Factor close to 1 (Unity).

Remember :
Good Power Factor
Saves Energy & Money

Stop Configuring MikroTik One by One.

Standardize. Automate. Scale.

OptiWize helps MSPs and system integrators manage all your MikroTik devices from one centralized platform — templates, backups, monitoring and more.



One Template.
Hundreds of Devices.



Automated Backups.
Always Protected.



Monitor & Troubleshoot.
Remotely.



Smart Alerts.
Instant Visibility.



Dashboard

Overview

- Devices
- Templates
- Backups
- Monitoring
- Alerts
- Troubleshooting
- Automation
- Marketplace
- Reports
- Settings

Devices

532

Online

Offline

12

Offline

Alerts

7

Alerts

Recent Backups

All devices

Backed up

2m ago

Template Applied

Office Branch 1

128 devices

30m ago

Automation

WMI Backup

Enabled

Devices Status



Online	532 (82%)
Offline	12 (2%)
Warning	24 (16%)

Recent Alerts

- Device offline
Branch Office 1 2m ago
- High CPU usage
Warehouse 3 10m ago
- Backup failed
Site Office 2 15m ago

Recent Activities

- Device added
Warehouse 2 1h ago
- Backup completed
Office Branch 1 2m ago
- Template applied
128 devices 30m ago



Save
Time



Reduce
Errors



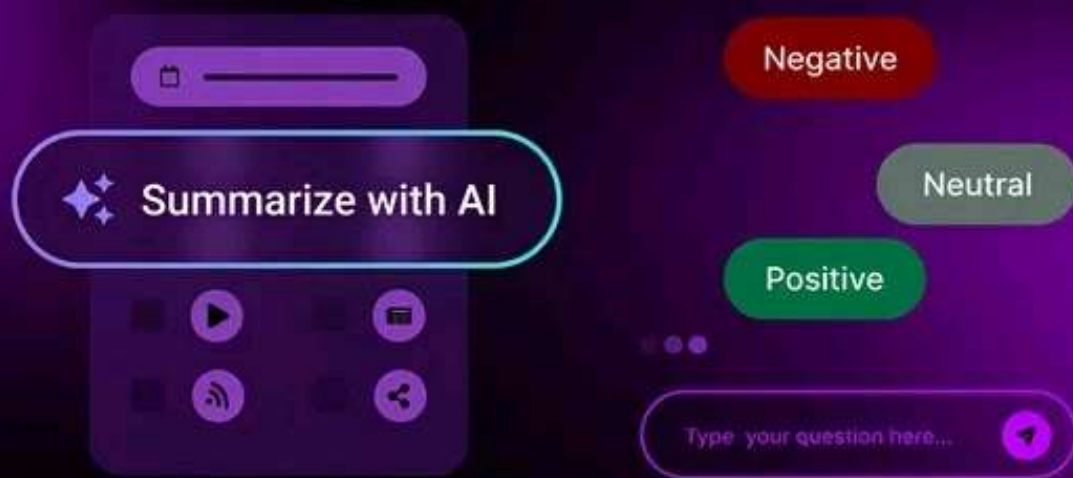
Scale With
Confidence



Secure &
Reliable



Deliver Better
for Your Clients



**#1 AI-powered internet
monitoring tool**

FINAL CALL

Grab your AI Certificate for \$20 — this offer dies tonight

AI Certificate Program

especially if you're over 40.

28 days. 15 minutes a day. In 1 week — you'll feel unstoppable.
In 2 weeks — you'll know more than 90% of your colleagues.
In 1 month — certificate.

Day 1



ChatGPT

Day 2



Gemini

Day 3



Claude

Day 4



Midjourney

Day 5



Synthesia

Day 6



Perplexity

Day 7



Canva Magic Studio

Day 8



DALL-E 3

Day 9



Jasper

Day 10



Notion AI

Day 11



Grok

Day 12



Copilot

Day 13



Fathom

Day 14



Zapier AI

Day 15



Beautiful.ai

SEE THE FULL PROGRAM

KENAPA PLN TIDAK MENYIMPAN LISTRIK SAJA?

KOTAK INFORMASI 1

LISTRIK HARUS DIGUNAKAN SAAT ITU JUGA

KOTAK INFORMASI 2

PASOKAN DAN PEMAKAIAN HARUS SELALU SEIMBANG



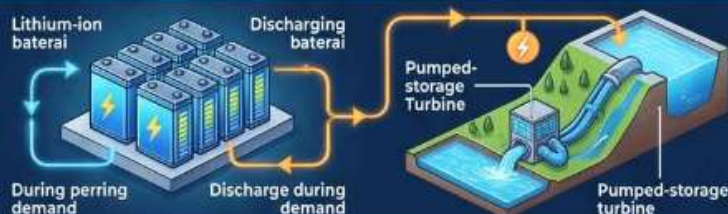
KOTAK INFORMASI 3

MENYIMPAN LISTRIK DALAM JUMLAH SANGAT BESAR TIDAK MUDAH



KOTAK INFORMASI 5

SEBAGIAN ENERGI DAPAT DISIMPAN MENGGUNAKAN BATERAI DAN PEMBANGKIT POMPAAN AIR



KOTAK FAKTA

SETIAP DETIK PLN HARUS MENYESUAIKAN PRODUKSI DENGAN KEBUTUHAN



KOTAK FAKTA

LISTRIK BERBEDA DENGAN AIR YANG BISA DISIMPAN DALAM WADUK



KESIMPULAN BESAR

LISTRIK BISA DISIMPAN, TETAPI BELUM SEMUDAH MENYIMPAN AIR



**THE WORLD'S ONLY
FREE AI CRYPTO ADVISOR.
NO CARD. NO CATCH.
NO BRAINER.**



SENSOR COMPARISON

INDUSTRIAL
SOLUTIONS

K-Type vs J-Type

K-Type Thermocouple



Temperature Range
-200°C to +1260°C



High Accuracy
±1.5°C or ±0.4%



Oxidation Resistant



Excellent Long-Term
Stability



Best for High
Temperature
Applications



J-Type Thermocouple



Temperature Range
-40°C to +750°C



Good Accuracy
±2.2°C or ±0.75%



Fast Thermal
Response



Ideal for Reducing
Atmospheres



Economical Choice

COMPARISON TABLE

PARAMETER	K-TYPE	J-TYPE
 Conductors	Chromel / Alumel	Iron / Constantan
 Temperature Range	-200°C to +1260°C	-40°C to +750°C
 Accuracy	±1.5°C	±2.2°C
 IEC Color Code	Yellow (+), Red (-)	Black (+), White (-)
 Corrosion Resistance	Excellent	Moderate
 Maximum Continuous Temperature	1260°C	750°C
 Typical Applications	Furnaces, Boilers, Turbines, Kilns	Ovens, Plastic Machines, HVAC, Food Industry



HIGH DURABILITY



WIDE TEMPERATURE
RANGE



FAST RESPONSE
TIME

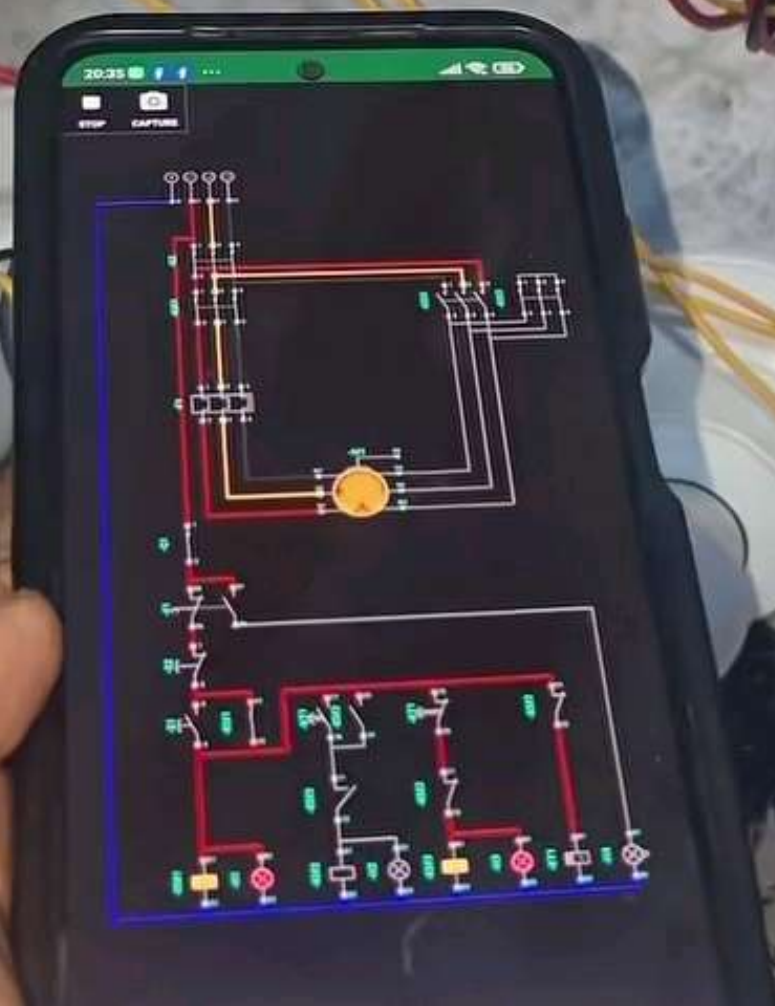


RELIABLE
PERFORMANCE



INDUSTRIAL
APPLICATIONS

Diys_tech





Electrical Knowledge



BASIC LEVEL DATA

1. BASIC UNITS (SI UNITS)

1. Voltage : Volt (V)
2. Current : Ampere (A)
3. Resistance : Ohm (Ω)
4. Power : Watt (W)
5. Energy : Kilowatt Hour (kWh)
6. Frequency : Hertz (Hz)
7. Capacitance : Farad (F)
8. Inductance : Henry (H)
9. Charge : Coulomb (C)
10. Time : Second (s)

2. BASIC CONVERSIONS

1. 1 kW = 1000 W
2. 1 MW = 1000 kW = 1,000,000 W
3. 1 HP = 746 W = 0.746 kW
4. 1 kW = 1.341 HP
5. 1 kVA = 1000 VA
6. 1 kWh = 1000 Wh (Unit)
7. 1 A = 1000 mA
8. 1 m = 1000 mm
9. 1 cm = 10 mm
10. 1 kg = 1000 g
11. 1 Minute = 60 Seconds
12. 1 Hour = 60 Minutes

3. POWER SYSTEM

1. Single Phase Voltage : 230 V (RMS)
2. Three Phase Voltage : 415 V (RMS)
3. Frequency (India) : 50 Hz
4. DC Voltage (Common) : 24 V, 48 V, 110 V
5. Battery Cell Voltage : 1.5 V (DC)
6. Standard Power Factor : 0.8 lag
7. Full Load Power Factor : 0.8 to 0.9
8. System Type : 1 Phase, 3 Phase
9. Earthing System : TT, TN, IT

4. WIRE & CABLE BASIC DATA

1. Copper Resistivity (at 20°C)
= $1.72 \times 10^{-8} \Omega \cdot m$
2. Aluminium Resistivity (at 20°C)
= $2.82 \times 10^{-8} \Omega \cdot m$
3. Standard Copper Density
= 8.96 g/cm^3
4. Standard Aluminium Density
= 2.70 g/cm^3
5. Allowable Current Density
(Copper) = 2 to 6 A/mm²
6. Allowable Current Density
(Aluminium) = 1.5 to 3.5 A/mm²

5. ELECTRICAL MATERIALS

1. Conductor : Copper, Aluminium
2. Insulator : PVC, XLPE, Rubber
3. Magnetic Material : Iron, Silicon Steel
4. Conductivity (Copper)
= $5.8 \times 10^7 \text{ S/m}$
5. Conductivity (Aluminium)
= $3.5 \times 10^7 \text{ S/m}$

6. BASIC ELECTRICAL VALUES

1. Ohm's Law : $V = I \times R$
2. Resistance (Ω) : V / I
3. Current (A) : V / R
4. Power (W) : $V \times I$
5. Power (W) : $I^2 \times R$
6. Power (W) : V^2 / R
7. Energy (Wh) : $P \times t$
(P = Power in W, t = Time in Hour)

7. COMMON ABBREVIATIONS

V : Volt	AC : Alternating Current
A : Ampere	PF : Power Factor
W : Watt	MCB : Miniature Circuit Breaker
kW : Kilowatt	MCCB : Moulded Case CB
kVA : Kilo Volt Ampere	RCCB : Residual Current CB
kWh : Kilo Watt Hour	TP : Three Phase
Hz : Hertz	SP : Single Phase
Ω : Ohm	NO : Normally Open
DC : Direct Current	NC : Normally Closed

8. SAFETY & STANDARDS

1. Electric Shock Voltage
> 50 V AC or > 120 V DC
is dangerous for human body.
2. Earthing Resistance (General)
 $\leq 1 \text{ Ohm}$ (Recommended)
3. Insulation Resistance (Minimum)
 $\geq 1 \text{ M}\Omega$ for Low Voltage System
4. Touch Voltage Limit (Safe)
 $\leq 50 \text{ V AC}$

QUICK NOTES

- ★ Always use correct units in calculations.
- ★ Higher the Power Factor, better the efficiency.
- ★ Use Copper for better conductivity.

- ★ Always follow Electrical Safety Rules.
- ★ Check earthing regularly.
- ★ Overloading reduces equipment life.

Remember

- Learn
- Understand
- Apply
- Succeed

⚡ POWER TRANSMISSION VOLTAGE LEVELS EXPLAINED ⚡

From Power Plant to Your Home



 TRANSMISSION DISTANCE	 RECOMMENDED VOLTAGE
 Up to 20 km	 11 kV
 20–50 km	 33 kV
 50–100 km	 66 kV
 100–200 km	 132 kV
 200–300 km	 220 kV
 300–500 km	 400 kV
 Above 500 km	 765 kV



NOTE:

Higher transmission voltages reduce current, minimize power losses, and improve long-distance transmission efficiency.

INSTALL ULANG LAPTOP & PC

2012



**LAPTOP/PC LEMOT,
SERING ERROR, VIRUS?
KAMI SOLUSINYA!**



**INSTALL ULANG WINDOWS
BERSIH & OPTIMAL**



**AMAN, STABIL,
PERFORMA MAKSIMAL**

SEMUA WINDOWS



XP



7



8.1



10



11

**BISA
INSTALL**



Photoshop



CorelDRAW



AutoCAD



SketchUp



SPSS



Office

**DAN
BANYAK
LAGI!**

Must-Know Ops Strategies for DevOps



CODING </>

DAPAT MEMBUAT ANAK ANDA
TAMPIL MENONJOL

PROGRAM CODING UNTUK **ANAK-ANAK**



UNTUK KELAS 1-12



TUTOR 1% TERBAIK



SIAP BERSAING DI MASA DEPAN



Transform Your **VMware Environment** with Microsoft Azure

One Trusted Partner for Your End-to-End VMware Transformation



Modernization
& Migration



Cost
Optimization



Security &
Compliance



Business Continuity
& Disaster Recovery



Hybrid Cloud



AI Readiness



Managed Services

UTC

18:09:24

Thu, May 21

LOCAL

14:09:24

Thu, May 21

DE - YOUR LOCATION

NOWRL

GRID

EM73uw

COORDS

33.9501°N, 84.2650°W

QTH

GA, United States

SUNRISE

06:39

SUNSET

20:34

DAYLIGHT

13h 54m

75°

F / C

HUMIDITY 78%

WIND 17 mph

Moderate Showers

CONTACTS & SPOTS - SHAPE - MODE - COLOR - BAND

CW

SSB

FT8/FT4

FM/AM

Digital

D-Star

160m

80m

60m

40m

20m

17m

15m

12m

DX NEWS

Active DXpeditions, current solar conditions, and NOAA alerts



Corona & Solar Wind (193 Å)

SFI

114

WIND

427 km/s

KP INDEX

1

BZ

-2 nT

PROPAGATION

SFI

114

K INDEX

2

A INDEX

8

INDUCTION MOTOR STARTERS

SUMMARY

An **INDUCTION MOTOR STARTER** is an electrical device used to **START, PROTECT** and **CONTROL** an induction motor safely. It limits **HIGH STARTING CURRENT (INRUSH CURRENT)**, protects against **OVERLOAD** and **LOW VOLTAGE**, and ensures **SMOOTH MOTOR OPERATION**.

TYPES OF INDUCTION MOTOR STARTERS

1



DOL STARTER (DIRECT-ON-LINE)

- Connects the motor directly to the full supply voltage.
- Ideal for **SMALL THREE-PHASE MOTORS** (usually rated **UNDER 5 kW**).

USE:

For small motors where high starting current is acceptable.

2



STAR-DELTA STARTER

- Starts the motor in **STAR** connection to reduce starting voltage and current.
- Switches to **DELTA** connection as the motor reaches 80-90% speed.

USE:

For medium to large three-phase squirrel cage motors.

3



AUTO-TRANSFORMER STARTER

- Uses a three-phase autotransformer to reduce the initial stator voltage.
- Gradually applies full voltage once the motor speeds up.

USE:

For large motors requiring high starting torque with reduced current.

4



SOFT STARTER

- Uses semiconductor devices (thyristors) to increase voltage gradually.
- Provides smooth, jerk-free starting and reduces mechanical stress.

USE:

For all types of motors where smooth starting is required.

5



ROTOR RESISTANCE STARTER

- Used only for **SLIP-RING INDUCTION MOTORS**.
- Adds external resistance in rotor circuit to lower starting current and increase starting torque.

USE:

For slip-ring motors needing high starting torque with low current.

KEY BENEFITS



LIMITS INRUSH CURRENT

Prevents high starting current damage.



PROTECTS MOTOR

Provides protection against overload, low voltage & faults.



REDUCES MECHANICAL STRESS

Ensures smooth acceleration and decreases wear and tear.



IMPROVES MOTOR LIFE AND RELIABILITY

Enhances performance and overall system reliability.

WHERE THEY ARE USED?



INDUSTRIAL
MACHINES



PUMPS &
COMPRESSORS



CONVEYORS



FANS &
BLOWERS



CRANES &
HOISTS



HVAC & OTHER
APPLICATIONS



Power the Future of DIGITAL ASSETS

Smart Infrastructure. Secure Rewards.

 Secure & Transparent

 Energy-Efficient
Infrastructure

 Global Network
Reliability



**GROW WITH
TECHNOLOGY.
EARN WITH
CONFIDENCE.**

Join a Global Community
Building the Future Together.



Advanced Security
Protocols



Distributed Network
Technology



Sustainable &
Eco-Friendly

MODEL: A2



Hashrate 226 TH/s
Power 3500 W
Energy Efficiency 15.5 J/TH
Noise Level ≤ 75 dB
Cooling System
Advanced Air Cooling



-  Real-time Monitoring
-  Automatic Reward
Distribution
-  Referral Rewards
Program
-  User-Friendly
Experience



CABLE CURRENT CARRYING CAPACITY

SAFE CURRENT THAT A CABLE CAN CARRY CONTINUOUSLY UNDER SPECIFIED CONDITIONS WITHOUT EXCEEDING ITS TEMPERATURE LIMIT.

1. DEFINITION

Current Carrying Capacity (CCC) or Ampacity is the maximum continuous current a cable can carry under standard conditions without exceeding its maximum permissible conductor temperature.

2. FACTORS AFFECTING CCC

- Conductor material (Copper / Aluminium)
- Conductor size (Cross-sectional area)
- Insulation type and temperature rating
- Installation method
- Ambient (air) temperature
- Number of loaded cores in the cable
- Grouping of cables (bunched installation)
- Soil thermal resistivity (for underground cables)
- Ventilation and other installation conditions

3. TYPICAL CABLE CONSTRUCTION

Conductor
(Copper / Aluminium)

Insulation
(PVC / XLPE)

Bedding
(Optional)

Armouring
(Steel Wires / Aluminium Wires)

Outer Sheath
(PVC / PE)



4. STANDARD REFERENCES

- IEC 60364-5-52
- IS 732 (Part 1)
- IEC 60287 (Calculation of Current Rating)
- National Electrical Code (NEC)

Note: Values may vary slightly between standards and manufacturers.

5. CURRENT CARRYING CAPACITY (CCC) TABLE

COPPER CONDUCTOR, PVC INSULATED & PVC SHEATHED CABLE (70°C), AMBIENT TEMP. 30°C

Nominal Cross-sectional Area (mm²)	Installation Method					
	In Conduit / Trunking (2 Loaded Conductors)	In Conduit / Trunking (3 Loaded Conductors)	Clipped Direct (On Wall) (2 Loaded Conductors)	Clipped Direct (On Wall) (3 Loaded Conductors)	Buried in Ground (Direct)	In Air (Free Hanging)
	Current Carrying Capacity (Ampere)					
1.5	18	15	20	16	24	26
2.5	24	20	27	21	32	36
4	32	27	36	28	41	47
6	41	34	46	36	53	61
10	57	47	63	50	72	83
16	76	63	85	67	94	108
25	101	84	114	90	120	138
35	125	104	141	111	145	167
50	151	126	171	134	170	196
70	192	160	218	171	201	232
95	232	193	264	207	232	268
120	269	224	306	241	261	301
150	310	259	353	278	292	337
185	353	295	402	316	320	369
240	415	346	472	371	362	418

6. TEMPERATURE CORRECTION FACTOR

Ambient Temperature (°C)	20	25	30	35	40	45	50	55	60
Correction Factor	1.12	1.06	1.00	0.94	0.87	0.79	0.71	0.61	0.50

Note: Multiply the table value by the correction factor for actual ambient temperature.

8. IMPORTANT NOTES

- Values are for reference only.
- Always derate for grouping, ambient temperature & installation.
- Ensure cable operating temperature does not exceed its rating (e.g., PVC 70°C, XLPE 90°C).
- Final selection should be verified as per the relevant standard and site conditions.

7. INSTALLATION METHODS (EXAMPLES)



In Conduit / Trunking



Clipped Direct (On Wall)



Buried in Ground



In Air (Free Hanging)

9. BASIC FORMULA (CALCULATION OF CCC)

$$CCC = I_z = I_t \times C_a \times C_g \times C_s$$

Where,

- I_t = Base current carrying capacity (from standard table)
- C_a = Ambient temperature correction factor
- C_g = Grouping correction factor
- C_s = Other correction factor (soil, installation, etc.)

Select cable such that:

$$I_b \leq I_n \leq I_z$$

- I_b = Design current
- I_n = Cable rating
- I_z = Corrected CCC

TIP: Always choose the next higher standard size if the calculated current is close to the cable's CCC.

ELECTRICAL CALCULATION

Next Level

By Electrical Knowledge



1 Transformer % Impedance

$S = 1000 \text{ kVA}$
 $V_L = 11 \text{ kV}$
 $P_{sc} = 12 \text{ kW}$
 $I_{sc} = ?$
 $Z\% = ?$

$$I_{FL} = \frac{S}{\sqrt{3} \times V_L}$$
$$I_{FL} = \frac{1000 \times 1000}{\sqrt{3} \times 11000} = 52.49 \text{ A}$$
$$Z\% = \frac{P_{sc} \times 100}{S}$$
$$Z\% = \frac{12 \times 100}{1000} = 1.2\%$$

$$I_{sc} = 52.49 \text{ A}, Z\% = 1.2\%$$

2 Maximum Demand

Connected Load = 500 kW
Demand Factor = 0.75
Diversity Factor = 0.8
Max. Demand = ?

$$\text{Max. Demand} = \text{Connected Load} \times \text{DF} \times \text{Diversity Factor}$$
$$= 500 \times 0.75 \times 0.8$$

$$\text{Max. Demand} = 300 \text{ kW}$$

3 Transformer Copper Loss

$I_{FL} = 50 \text{ A}$
 $R_{eq} = 0.15 \Omega$
 $P_{cu} = ? (3\phi)$

$$P_{cu} = 3 \times I_{FL}^2 \times R_{eq}$$
$$= 3 \times 50^2 \times 0.15$$
$$= 1125 \text{ W}$$

$$P_{cu} = 1.125 \text{ kW}$$

4 Transformer Iron Loss

No Load Current = 5 A
 $V_L = 11 \text{ kV}$
 $P_0 = ?$

$$I_w = I_0 \times \cos \phi_0 \quad (\text{assume } \cos \phi_0 = 0.2)$$
$$I_w = 5 \times 0.2 = 1 \text{ A}$$
$$P_0 = \sqrt{3} \times V_L \times I_w$$
$$= 1.732 \times 11000 \times 1$$

$$P_0 = 19.05 \text{ kW}$$

5 Voltage Drop in 3Φ Line (General)

$I = 150 \text{ A}$
 $R = 0.05 \Omega/\text{km}$
 $X = 0.08 \Omega/\text{km}$
 $L = 120 \text{ m}$
 $\cos \phi = 0.8 \text{ (lag)}$
 $V_D = ?$

$$V_D = \frac{\sqrt{3} \times I \times L \times (R \cos \phi + X \sin \phi)}{1000}$$
$$V_D = \frac{2.732 \times 150 \times 0.12 \times (0.05 \times 0.8 + 0.08 \times 0.6)}{1000}$$

$$V_D = 2.52 \text{ V}$$

6 Motor Efficiency

Input Power = 25 kW
Output Power = 22 kW
 $\eta = ?$

$$\eta = \frac{\text{Output}}{\text{Input}} \times 100$$
$$= \frac{22}{25} \times 100$$

$$\eta = 88\%$$

7 Fault Level at Busbar

$S_{sc} = 500 \text{ MVA}$
 $V = 11 \text{ kV}$
 $I_{sc} = ?$

$$I_{sc} = \frac{S_{sc} \times 1000}{\sqrt{3} \times V}$$
$$I_{sc} = \frac{500 \times 1000}{1.732 \times 11}$$

$$I_{sc} = 26.24 \text{ kA}$$

8 CT Burden

CT Ratio = 1000 / 5
Burden per VA = 10 VA
Total Burden = ?

$$\text{Total Burden} = \text{Burden per VA} \times \text{Sec. Current}$$
$$= 10 \times 5$$

$$\text{Total Burden} = 50 \text{ VA}$$

9 Cable Current Carrying Capacity

Cable Size = 95 mm²
Installation = Direct in Air
From Standard Table
Current Rating = ?

From Standard Table (Cu Cable)
95 mm² (Direct in Air)

$$\text{Current Rating} = 290 \text{ A}$$

10 Capacitor kVAR (3Φ Bank)

$C = 50 \mu\text{F}$ (per phase)
 $V_L = 415 \text{ V}$
 $f = 50 \text{ Hz}$
kVAR = ?

$$\text{kVAR} = \frac{\sqrt{3} \times V_L^2 \times 2\pi f C}{1000}$$
$$= \frac{1.732 \times 415^2 \times 2 \times 3.14 \times 50 \times 10^{-6}}{1000}$$

$$\text{kVAR} = 5.89 \text{ kVAR}$$

11 Energy & Cost

Load = 60 kW
Time = 18 hr/day
Days = 30
Rate = ₹ 7 /kWh
Cost = ?

$$\text{Energy} = 60 \times 18 \times 30 = 32400 \text{ kWh}$$
$$\text{Cost} = \text{Energy} \times \text{Rate}$$
$$= 32400 \times 7$$

$$\text{Cost} = ₹ 2,26,800$$

12 Power Factor from Readings

Wattmeter Reading = 40 kW
Voltmeter Reading = 415 V
Ammeter Reading = 80 A
PF = ? (3Φ)

$$P = \sqrt{3} \times V \times I \times \text{PF}$$
$$\text{PF} = \frac{P}{\sqrt{3} \times V \times I}$$
$$= \frac{40000}{1.732 \times 415 \times 80}$$

$$\text{PF} = 0.697 \text{ (lag)}$$

13 Wattless Current

$I = 100 \text{ A}$
PF = 0.6 (lag)
Wattless Current (I_w) = ?

$$I_w = I \times \sin (\cos^{-1} \text{PF})$$
$$= 100 \times \sqrt{1 - 0.6^2}$$
$$= 100 \times 0.8$$

$$I_w = 80 \text{ A}$$

14 Star-Delta Motor Starting Current

DOL Starting Current = 300 A
Starting Current in Star = ?

$$I_{star} = \frac{I_{DOL}}{3}$$
$$= \frac{300}{3}$$

$$I_{star} = 100 \text{ A}$$

15 Conductor Resistance (Given Resistivity)

$\rho = 0.0175 \Omega \cdot \text{mm}^2/\text{m}$
 $L = 200 \text{ m}$
 $A = 50 \text{ mm}^2$
 $R = ?$

$$R = \frac{\rho \times L}{A}$$
$$= \frac{0.0175 \times 200}{50}$$

$$R = 0.07 \Omega$$

16 kWh to kVAh Conversion

Energy = 500 kWh
PF = 0.8
kVAh = ?

$$\text{kVAh} = \frac{\text{kWh}}{\text{PF}}$$
$$= \frac{500}{0.8}$$

$$\text{kVAh} = 625 \text{ kVAh}$$



Learn • Understand • Apply • Excel



ELECTRICAL CALCULATION



By Electrical Knowledge

Next Level

1 Power from 3 Phase Line Values

Given :

- $V_L = 415 \text{ V}$
- $I_L = 18 \text{ A}$
- $\text{PF} = 0.85$
- $P = ?$



Formula & Calculation :

$$P = \sqrt{3} \times V_L \times I_L \times \text{PF}$$

$$P = 1.732 \times 415 \times 18 \times 0.85$$

$$P = 1.732 \times 7470 \times 0.85$$

$$P = 12930.24 \times 0.85$$

$$P = 10990.70 \text{ W}$$

$$P = 10.99 \text{ kW}$$

2 Line Current from Power (3 Phase)

Given :

- $P = 15 \text{ kW}$
- $V_L = 440 \text{ V}$
- $\text{PF} = 0.90$
- $I_L = ?$



Formula & Calculation :

$$I_L = P / (\sqrt{3} \times V_L \times \text{PF})$$

$$I_L = 15000 / (1.732 \times 440 \times 0.90)$$

$$I_L = 15000 / 685.87$$

$$I_L = 21.87 \text{ A}$$

$$I_L = 21.87 \text{ A}$$

3 Power Factor from Real & Apparent Power

Given :

- $P = 12 \text{ kW}$
- $S = 16 \text{ kVA}$
- $\text{PF} = ?$



Formula & Calculation :

$$\text{PF} = P / S$$

$$\text{PF} = 12 / 16$$

$$\text{PF} = 0.75$$

$$\text{PF} = 0.75$$

4 Apparent Power from Real Power & PF

Given :

- $P = 9 \text{ kW}$
- $\text{PF} = 0.60$
- $S = ?$



Formula & Calculation :

$$S = P / \text{PF}$$

$$S = 9 / 0.60$$

$$S = 15 \text{ kVA}$$

$$S = 15 \text{ kVA}$$

5 Reactive Power (kVAR) from Real Power & PF

Given :

- $P = 10 \text{ kW}$
- $\text{PF} = 0.80$
- $Q = ?$



Formula & Calculation :

$$Q = P \times \tan(\cos^{-1} \text{PF})$$

$$Q = 10 \times \tan(\cos^{-1} 0.80)$$

$$Q = 10 \times \tan(36.87^\circ)$$

$$Q = 10 \times 0.75$$

$$Q = 7.5 \text{ kVAR}$$

$$Q = 7.5 \text{ kVAR}$$

6 Voltage Drop in 3 Phase Line (Approx.)

Given :

- $I_L = 25 \text{ A}$
- $R = 0.4 \Omega/\text{km}$
- $X = 0.3 \Omega/\text{km}$
- $L = 100 \text{ m (0.1 km)}$
- $V_L = 415 \text{ V}$
- $\text{VD} = ?$



Formula & Calculation :

$$\text{VD} = \sqrt{3} \times I_L \times (R \cos\phi + X \sin\phi) \times L$$

Assume $\text{PF} = 0.8$ ($\cos\phi = 0.8$, $\sin\phi = 0.6$)

$$\text{VD} = 1.732 \times 25 \times (0.4 \times 0.8 + 0.3 \times 0.6) \times 0.1$$

$$\text{VD} = 1.732 \times 25 \times (0.32 + 0.18) \times 0.1$$

$$\text{VD} = 1.732 \times 25 \times 0.50 \times 0.1$$

$$\text{VD} = 2.165 \text{ V}$$

$$\text{VD} = 2.17 \text{ V (Approx.)}$$

7 Energy Cost Calculation

Given :

- $P = 7.5 \text{ kW}$
- $t = 10 \text{ h/day}$
- $\text{Days} = 30$
- $\text{Rate} = ₹ 8 / \text{kWh}$
- $\text{Cost} = ?$



Formula & Calculation :

$$E = P \times t \times \text{Days}$$

$$E = 7.5 \times 10 \times 30$$

$$E = 2250 \text{ kWh}$$

$$\text{Cost} = E \times \text{Rate}$$

$$\text{Cost} = 2250 \times 8$$

$$\text{Cost} = ₹ 18,000$$

$$\text{Cost} = ₹ 18,000$$

8 Motor Efficiency Calculation

Given :

- $\text{Input Power} = 11 \text{ kW}$
- $\text{Output Power} = 9.2 \text{ kW}$
- $\eta = ?$



Formula & Calculation :

$$\eta = \frac{\text{Output Power}}{\text{Input Power}} \times 100\%$$

$$\eta = 9.2 / 11 \times 100\%$$

$$\eta = 0.836 \times 100\%$$

$$\eta = 83.64\%$$

$$\eta = 83.64\%$$

Important Formulas

- $P = V \times I$ (Single Phase)
- $P = \sqrt{3} \times V_L \times I_L \times \text{PF}$ (3 Phase)
- $I_L = P / (\sqrt{3} \times V_L \times \text{PF})$

- $S = P / \text{PF}$
- $Q = \sqrt{S^2 - P^2}$
- $Q = P \times \tan(\cos^{-1} \text{PF})$
- $E (\text{kWh}) = P (\text{kW}) \times t (\text{h})$



Learn • Understand • Apply • Excel

Pelatihan AI Gratis

**MAJU
BARENG** 

Part of the AI Opportunity Fund: Asia-Pacific

In collaboration with 

with support from  

AI Productivity and AI API Integration for Developers

Sesi 1

27 Juli

Membangun dan mengembangkan website dengan bantuan AI untuk mempercepat proses dan meningkatkan kualitas fitur.

Sesi 2

29 Juli

Mengimplementasikan Gemini AI API multimodal untuk memproses teks dan gambar dalam aplikasi berbasis AI.

Sesi 3

31 Juli

Mengembangkan chatbot cerdas berbasis Gemini AI Model yang mampu memahami konteks dan merespons secara natural.

Benefit

-  3 Sesi pembelajaran
-  Sertifikat resmi
-  Online via Zoom

00.01



1 perangkat

0.00 KB/dtk 4G

Mode data

Beli data

facebook



Pelatihan AI Gratis

MAJU BARENG

Part of the AI Opportunity Fund: Asia-Pacific

In collaboration with **avpn**with support from **Google.org** **ADB**

AI Productivity and AI API Integration for Developers

Sesi 1

27 Juli

Membangun dan mengembangkan website dengan bantuan AI untuk mempercepat proses dan meningkatkan kualitas fitur.

Sesi 2

29 Juli

Mengimplementasikan Gemini AI API multimodal untuk memproses teks dan gambar dalam aplikasi berbasis AI.

Sesi 3

31 Juli

Mengembangkan chatbot cerdas berbasis Gemini AI Model yang mampu memahami konteks dan merespons secara natural.

Benefit

3 Sesi pembelajaran

Sertifikat resmi

Online via Zoom



Terbuka untuk ICT workers

Daftar kelas terdekat sebelum terlambat:

my.hacktiv8.com/majubarengai26

Kelas Gratis untuk IT Professional

Pelajari selengkapnya



240



24



20

**KOMUNITAS RAKIT PC KE...**

Denta · 4 hr ·

dari nada na kenake

ELECTRICAL CALCULATION

ADVANCED Level

By Electrical Knowledge



1 Transformer Efficiency

Output = 800 kW
Voltage = 11 kV
Current = 50 A
Power Factor = 0.85
Iron Loss = 6 kW
Copper Loss = 8 kW
Efficiency = ?

$$P_{out} = \sqrt{3} \times V \times I \times PF$$

$$= \sqrt{3} \times 11 \times 10^3 \times 50 \times 0.85$$

$$= 809.15 \text{ kW}$$

$$\text{Total Loss} = \text{Iron Loss} + \text{Copper Loss}$$

$$= 6 + 8 = 14 \text{ kW}$$

$$\eta = \frac{P_{out}}{P_{out} + \text{Loss}} \times 100$$

$$= \frac{809.15}{809.15 + 14} \times 100$$

$$\eta = 98.29 \%$$

2 ATR for Transformer

HV Turns = 11000
LV Turns = 415
Measured Ratio = 26.10
Actual Ratio = ?
% Error = ?

$$\text{Actual Ratio} = \frac{\text{HV Turns}}{\text{LV Turns}}$$

$$= \frac{11000}{415} = 26.51$$

$$\% \text{ Error} = \frac{\text{Actual} - \text{Measured}}{\text{Actual}} \times 100$$

$$= \frac{26.51 - 26.10}{26.51} \times 100$$

$$\text{Actual Ratio} = 26.51, \% \text{ Error} = 1.55\%$$

3 Short Circuit Current (MVA Method)

S = 750 MVA
V = 33 kV
Z% = 8 %
I_{sc} = ?

$$I_{sc} = \frac{S \times 100}{\sqrt{3} \times V \times Z\%}$$

$$= \frac{750 \times 100}{\sqrt{3} \times 33 \times 8}$$

$$= \frac{75000}{457.11}$$

$$I_{sc} = 164.07 \text{ A}$$

4 Load Current from kW & PF (3φ)

P = 120 kW
V = 415 V
PF = 0.9
I = ?

$$I = \frac{P}{\sqrt{3} \times V \times PF}$$

$$= \frac{120000}{\sqrt{3} \times 415 \times 0.9}$$

$$= \frac{120000}{648.34}$$

$$I = 184.98 \text{ A}$$

5 Voltage Drop (3φ) Line

I = 200 A
L = 200 m
R = 0.07 Ω/km
X = 0.08 Ω/km
cosφ = 0.8
V_D = ?

$$V_D = \frac{\sqrt{3} \times I \times L \times (R \cos\phi + X \sin\phi)}{1000}$$

$$= \frac{\sqrt{3} \times 200 \times 0.2 \times (0.07 \times 0.8 + 0.08 \times 0.6)}{1000}$$

$$= \frac{69.28}{1000}$$

$$V_D = 6.93 \text{ V}$$

6 kVA from kW & PF

P = 150 kW
PF = 0.75
S (kVA) = ?

$$S = \frac{P}{PF}$$

$$= \frac{150}{0.75}$$

$$S = 200 \text{ kVA}$$

7 Capacitor Rating for PF Improvement

P = 100 kW
PF₁ = 0.6 (lag)
PF₂ = 0.95 (lag)
kVAR = ?

$$Q_c = P (\tan \cos^{-1} PF_1 - \tan \cos^{-1} PF_2)$$

$$= 100 (\tan 53.13^\circ - \tan 18.19^\circ)$$

$$= 100 (1.333 - 0.3287)$$

$$Q_c = 100.43 \text{ kVAR}$$

8 Energy Cost (Monthly)

Energy = 4500 kWh
Rate = ₹ 8 / kWh
PF Incentive = 2%
Cost = ?

$$\text{Base Cost} = 4500 \times 8 = 36000$$

$$\text{PF Incentive} = 2\% \text{ of } 36000 = 720$$

$$\text{Net Cost} = 36000 - 720$$

$$\text{Cost} = ₹ 35,280$$

9 Cable Size from Current Density

I = 160 A
J (Current Density) = 2.5 A/mm²
Area (A) = ?

$$A = \frac{I}{J} = \frac{160}{2.5}$$

$$A = 64 \text{ mm}^2$$

10 Conductor Resistance

p = 0.0175 Ω·mm²/m
L = 150 m
A = 70 mm²
R = ?

$$R = \frac{p \times L}{A} = \frac{0.0175 \times 150}{70}$$

$$R = 0.0375 \Omega$$

11 Inductive Reactance

f = 50 Hz
L = 0.15 H
X_L = ?

$$X_L = 2 \pi f L = 2 \pi \times 50 \times 0.15$$

$$X_L = 47.12 \Omega$$

12 Capacitive Reactance

f = 50 Hz
C = 80 μF
X_C = ?

$$X_C = \frac{1}{2 \pi f C} = \frac{1}{2 \pi \times 50 \times 80 \times 10^{-6}}$$

$$X_C = 39.79 \Omega$$

13 Star-Delta Impedance Conversion

Star Impedance (each) = 6 Ω
Delta Impedance (each) = ?

$$Z_\Delta = 3 \times Z_Y = 3 \times 6$$

$$Z_\Delta = 18 \Omega$$

14 Delta-Star Impedance Conversion

Delta Impedance (each) = 15 Ω
Star Impedance (each) = ?

$$Z_Y = \frac{Z_\Delta}{3} = \frac{15}{3}$$

$$Z_Y = 5 \Omega$$

15 Three Phase Active Power

V_L = 11 kV
I_L = 120 A
PF = 0.85
P = ?

$$P = \sqrt{3} \times V_L \times I_L \times PF$$

$$= \sqrt{3} \times 11000 \times 120 \times 0.85$$

$$= 1,945,542 \text{ W}$$

$$P = 1945.54 \text{ kW}$$

16 Average Current (3φ)

Energy = 7200 kWh
Voltage = 415 V
PF = 0.9
Time = 24 hr
I_{avg} = ?

$$I_{avg} = \frac{\text{Energy}}{\sqrt{3} \times V \times PF \times \text{Time}} \times 1000$$

$$= \frac{7200}{\sqrt{3} \times 415 \times 0.9 \times 24} \times 1000$$

$$I_{avg} = 66.27 \text{ A}$$



Learn • Understand • Apply • Excel





STATIC HEAD vs RESIDUAL PRESSURE CALCULATION



KEY COMPONENTS OF TOTAL DYNAMIC HEAD (TDH)

1. STATIC HEAD (H_s)

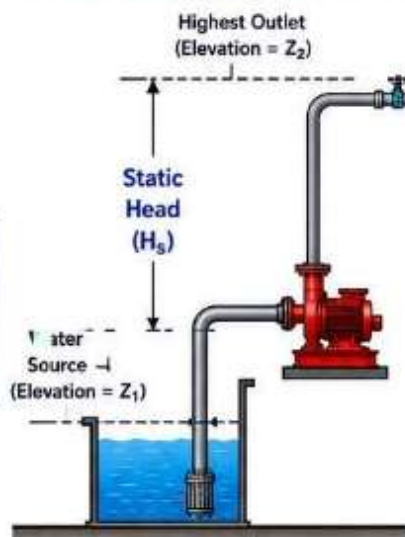
The vertical distance between the water level in the source (suction water level) and the highest point of discharge (outlet).

$$H_s = Z_2 - Z_1$$

Where,

Z_2 = Elevation of highest outlet (m)

Z_1 = Elevation of water level in source (m)



2. RESIDUAL PRESSURE (P_r)

The minimum pressure required at the most remote outlet (hydrant/sprinkler) to ensure effective operation.

$$H_r = \frac{P_r}{\rho \times g}$$

Where,

P_r = Required residual pressure (Pa or bar)

ρ = Density of water (1000 kg/m³)

g = Acceleration due to gravity (9.81 m/s²)

TYPICAL DESIGN VALUES (NFPA 20)

System Type	Minimum Residual Pressure
Sprinkler System	0.7 bar (10 psi)
Standpipe System	2.1 bar (30 psi)
Hydrant System	2.8 bar (40 psi)

TOTAL DYNAMIC HEAD (TDH)

$$\text{TDH} = \text{Static Head (H}_s\text{)} + \text{Friction Loss (H}_f\text{)} + \text{Residual Pressure Head (H}_r\text{)}$$

(All heads in meters of water column)

3. STATIC HEAD CALCULATION - EXAMPLE

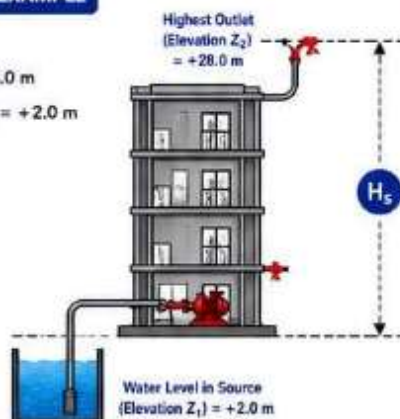
Given:

- Elevation of highest outlet (Z_2) = +28.0 m
- Elevation of water level in source (Z_1) = +2.0 m

Calculation:

$$\begin{aligned} H_s &= Z_2 - Z_1 \\ &= 28.0 - 2.0 \\ &= 26.0 \text{ m} \end{aligned}$$

Static Head (H_s) = 26.0 m



4. RESIDUAL PRESSURE CALCULATION - EXAMPLE

Given:

- Required residual pressure at most remote hydrant = 2.8 bar (Hydrant System as per NFPA 20)

Calculation:

$$\begin{aligned} H_r &= \frac{P_r}{\rho \times g} \\ &= \frac{2.8 \times 10^5}{1000 \times 9.81} \\ &= 28.54 \text{ m} \end{aligned}$$

Residual Pressure Head (H_r) = 28.5 m



5. SUMMARY OF HEAD COMPONENTS

Component	Symbol	Value (m)
Static Head	H_s	26.0
Friction Loss in Pipes & Fittings	H_f	15.0
Residual Pressure Head	H_r	28.5
TOTAL DYNAMIC HEAD (TDH)	TDH	69.5 m

6. TDH CALCULATION

$$\begin{aligned} \text{TDH} &= H_s + H_f + H_r \\ &= 26.0 + 15.0 + 28.5 \end{aligned}$$

$$\text{TDH} = 69.5 \text{ m}$$

DEFINITION OF TERMS

- H_s = Static Head (m)
- H_f = Friction Loss (m)
- H_r = Residual Pressure Head (m)
- TDH = Total Dynamic Head (m)
- P_r = Residual Pressure (bar or psi)

7. DESIGN TIPS



- Always include all static lift (positive or negative).
- Ensure residual pressure meets NFPA 20 minimums.
- Add adequate safety margin (10–15%) on TDH.
- Verify pump performance on the pump curve at design flow.
- Ensure NPSH available is greater than NPSH required.

8. QUICK REFERENCE - CONVERSION

PRESSURE TO HEAD

$$H_r \text{ (m)} = P_r \text{ (bar)} \times 10.197$$

Examples:

- 0.7 bar = 7.14 m
- 2.1 bar = 21.41 m
- 2.8 bar = 28.55 m

9. COMMON RESIDUAL PRESSURE

System	Min. Residual Pressure	
	bar	psi
Sprinkler	0.7	10
Standpipe	2.1	30
Hydrant	2.8	40



DESIGN GOAL:

Right Head Calculation → Right Pump Selection → Reliable Fire Protection



NFPA 20 COMPLIANT

ELECTRICAL ABBREVIATIONS & SYMBOLS

• AC	— Alternating Current	
• DC	— Direct Current	
• V	— Voltage	
• A	— Ampere	
• W	— Watt	
• kW	— Kilowatt	
• MW	— Megawatt	
• kVA	— Kilovolt Ampere	
• kVAR	— Kilovolt Reactive	
• HP	— Horsepower	
• Hz	— Hertz	
• PF	— Power Factor	
• MCB	— Miniature Circuit Breaker	
• MCCB	— Molded Case Circuit Breaker	
• RCCB	— Residual Current Circuit Breaker	
• ELCB	— Earth Leakage Circuit Breaker	
• RCBO	— Residual Current Breaker with Overcurrent Protection	
• SPD	— Surge Protection Device	
• CT	— Current Transformer	
• PT	— Potential Transformer	
• LT	— Low Tension	
• HT	— High Tension	

COMMON ELECTRICAL UNITS

 Voltage	Volt (V)	 Frequency	Hertz (Hz)
 Current	Ampere (A)	 Power Factor	PF
 Power	Watt (W)	 Resistance	Ohm (Ω)
 Energy	Kilowatt-hour (kWh)		



Digital Multimeter



Single Pole MCB



Double Pole MCB



RCCB



RCBO



MCCB



Surge Protection Device (SPD)



Current Transformer (CT)



Modular Contactor





VS

POWER USERS USE

CLAUDE to build systems



Research

Deeper insights,
faster



Documents

Analyze, extract,
understand



Workflows

Automate processes,
save time



AI Agents

Build your own
AI team



Strategy

Think bigger,
make better decisions



DISC INSULATOR CALCULATION

ENSURING SAFE INSULATION & RELIABLE POWER TRANSMISSION

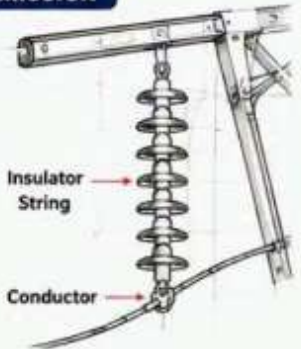


WHAT IS A DISC INSULATOR?

Disc insulators are used in overhead transmission lines to mechanically support conductors and electrically insulate them from the ground or tower.

KEY FUNCTIONS

- ✓ Provide mechanical support
- ✓ Provide electrical insulation
- ✓ Withstand mechanical & electrical stresses
- ✓ Ensure safety and reliability in power system



CALCULATION FORMULA

Number of Disc Insulators (n)

$$n = \frac{V_{sys} \times K_s}{V_d}$$

Where,

n = Number of disc insulators

V_{sys} = System voltage (kV)

K_s = Safety factor (usually 1.2 to 1.5)

V_d = Voltage withstand per disc (kV)

EXAMPLE

System Voltage (V_{sys}) = 132 kV

Voltage per Disc (V_d) = 11 kV

Safety Factor (K_s) = 1.3

$$n = \frac{132 \times 1.3}{11}$$

$$n = 15.6$$

Therefore,

Provide 16 Disc Insulators

VOLTAGE DISTRIBUTION ACROSS DISC



The voltage across each disc is not uniform.

The voltage across the unit nearest to conductor is highest.

The voltage across units decreases gradually towards the earthed end.

Therefore, string efficiency is always less than 100%.

IMPORTANT FACTORS



SYSTEM VOLTAGE (kV)

Higher the voltage, more number of discs are required.



VOLTAGE PER DISC

Depends on the type, size and design of disc insulator.



SAFETY FACTOR

Generally taken 1.2 to 1.5 to ensure safety under all conditions.



POLLUTION LEVEL

In polluted areas, longer creepage distance is required.



CREEPAGE DISTANCE

Minimum creepage distance must be maintained as per standards (IEC/IS).



MECHANICAL STRENGTH

Must withstand mechanical loads due to conductor weight, wind, ice, and fault conditions.

TYPES OF DISC INSULATORS

STANDARD DISC INSULATOR



Used in normal conditions.

FOG TYPE DISC INSULATOR



Used in heavy polluted areas.

ANTI-FOG TYPE DISC INSULATOR



Better performance in very heavy pollution.

DISC INSULATOR STRING CONFIGURATION

SUSPENSION STRING



Used in straight line supports.

TENSION STRING



Used in angle or corner supports.

DOUBLE STRING



Used in extra high voltage lines.

BENEFITS

- ✓ High mechanical strength
- ✓ Excellent electrical insulation
- ✓ Easy to replace
- ✓ Long service life
- ✓ Cost effective

MDB (MAIN DISTRIBUTION BOARD) PANEL



Distributes power from incoming supply to various sub distribution boards.



Ensures protection with MCCB/ACB, Fuses & Relays.



Provides metering & monitoring of electrical parameters.



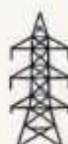
Designed for safety, reliability & easy maintenance.



STANDARDS REFERENCE



IS 2739 (Part 1) – Specification for Ceramic & Glass Insulators
IEC 60305 – Power Insulators
IEC 60815 – Selection & Dimensioning Recommendations



APPLICATION

- ✓ Overhead Transmission Lines
- ✓ Substation Equipment
- ✓ Railway Electrification
- ✓ Industrial Power Systems

CONCLUSION



Proper calculation and selection of disc insulators ensures the safety, reliability and efficiency of power transmission systems. Always follow standards and consider all environmental and mechanical factors.

Integrated Data Center Infrastructure

Built for AI, edge, and
high-density computing.



Power



Cooling



IT Space



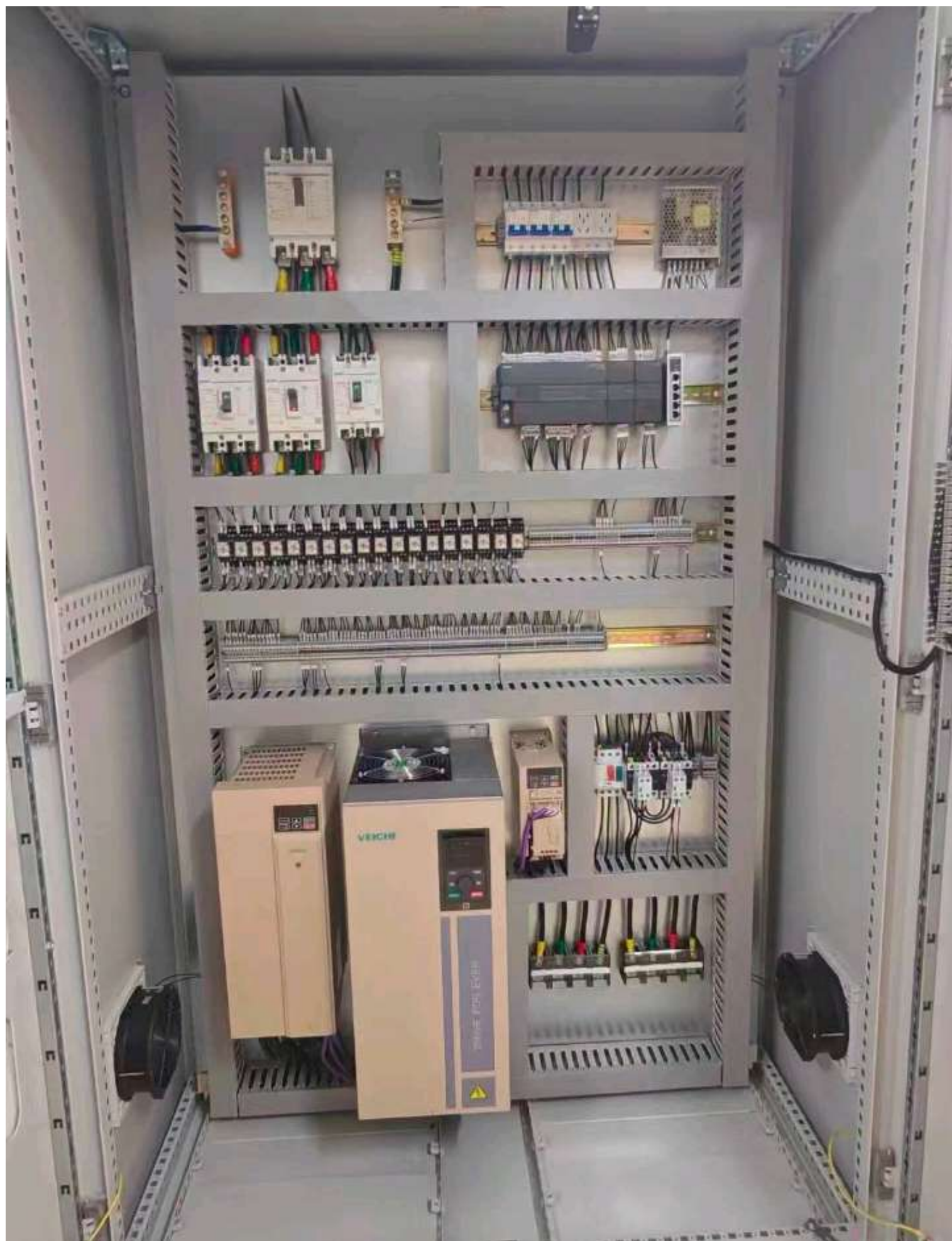
Monitoring



Fire Protection



Security





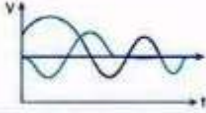
VFD SIZING CALCULATION



SELECT RIGHT VFD, GET MAXIMUM EFFICIENCY

WHAT IS VFD?

A Variable Frequency Drive (VFD) controls the speed of an AC motor by varying the frequency and voltage supplied to the motor.



WHY PROPER VFD SIZING?

- ✓ Prevents overloading and tripping
- ✓ Ensures long equipment life
- ✓ Better energy efficiency
- ✓ Optimizes system performance
- ✓ Avoids unnecessary extra cost



INFORMATION REQUIRED

- Motor Full Load Power (kW or HP)
- Motor Full Load Current (FLA)
- Motor Voltage (V)
- Motor Frequency (Hz)
- Application Type (Normal Duty / Heavy Duty)
- Service Factor (SF) – if any
- Ambient Temperature

SERVICE FACTOR GUIDE

APPLICATION TYPE	SERVICE FACTOR (SF)
Normal Duty	1.15
Heavy Duty	1.20
High Inertia / Shock Load	1.25

VFD SELECTION CHECKLIST

- ✓ Voltage Compatibility
- ✓ Current Rating
- ✓ Overload Capacity
- ✓ Ambient Temperature Rating
- ✓ Protection Class (IP Rating)
- ✓ Built-in Features
- ✓ Manufacturer Recommendation

BASIC SIZING FORMULA

1. Based on Motor Power

$$\text{VFD Power Rating (kW)} \geq \text{Motor Power (kW)} \times \text{Service Factor}$$

2. Based on Motor Current

$$\text{VFD Current Rating (A)} \geq \text{Motor FLA (A)} \times \text{Service Factor}$$

Where,

- Service Factor (SF) = 1.15 to 1.25 (Recommended)
- Select next higher standard VFD rating available

STEP-BY-STEP CALCULATION

1 Find Motor Full Load Power (kW)

$$\text{Motor Power (kW)} = \text{Motor HP} \times 0.746$$

2 Apply Service Factor (SF)

$$\text{Required VFD Power (kW)} = \text{Motor Power} \times \text{SF}$$

3 Check Motor Full Load Current (FLA)

$$\text{Required VFD Current (A)} = \text{Motor FLA} \times \text{SF}$$

4 Select VFD

Choose VFD with power and current rating greater than or equal to calculated value.

TIPS

- For multiple motors, size VFD for the largest motor.
- Ensure VFD has enough overload capacity (typically 150% for 60 sec).
- Always check manufacturer datasheet.

EXAMPLE CALCULATION

Given:

- Motor Power = 30 HP
- Motor Voltage = 400 V, 3 Phase
- Motor FLA = 52 A
- Frequency = 50 Hz
- Application = Normal Duty
- Service Factor (SF) = 1.15

1 Convert Motor Power to kW

$$\text{Motor Power (kW)} = 30 \times 0.746 = 22.38 \text{ kW}$$

2 Required VFD Power

$$= 22.38 \times 1.15 = 25.74 \text{ kW}$$

3 Required VFD Current

$$= 52 \times 1.15 = 59.8 \text{ A}$$

4 Select VFD

Select next higher standard rating.
Choose VFD with rating $\geq 26 \text{ kW}$ and $\geq 60 \text{ A}$.

Recommended VFD: 30 kW (or 37 kW), 400 V, 3 Phase, 60 A or higher

COMMON VFD RATING TABLE (EXAMPLE)

VFD POWER (kW)	VFD POWER (HP)	VFD CURRENT (A) @ 400V
7.5	10	17
11	15	25
15	20	32
18.5	25	40
22	30	50
30	40	60
37	50	75
45	60	90
55	75	110
75	100	150

Note: Ratings are indicative. Check manufacturer datasheet for exact values.

OVERLOAD CAPACITY (TYPICAL)



150% for 60 seconds
200% for 3 seconds

IMPORTANT NOTES

- Never select a VFD smaller than motor rating.
- Oversizing VFD by 1 step is recommended.
- Consider harmonics and derating for high ambient.
- Use proper input protection and grounding.

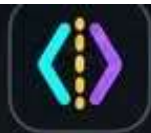


BENEFITS OF RIGHT SIZING

- ✓ Longer motor & VFD life
- ✓ Reduced breakdown & maintenance
- ✓ Better energy saving
- ✓ Smooth & reliable operation



RIGHT VFD SIZE = RELIABILITY + EFFICIENCY + SAVINGS



RiftMerge



See What's Really Different.

Chat + Claude + Gemini

First principles say one AI workspace
beats a pile of subscriptions.



●●● bash - root@jkt-2

```
$ ping vps-lama.io
```

```
Request timeout for icmp_seq 0
```

```
Request timeout for icmp_seq 1
```

```
$ ssh root@jkt-2.nevacloud.com
```

```
Connected! Latency: 0.4ms - Jakarta DC
```

```
AMD EPYC - NVMe 8Gb/s - 3x replication
```

```
$ __
```

VPS-mu timeout.

Saatnya **upgrade.**

Server lokal Jakarta. Deploy dalam 50 detik. Tanpa egress fee.

AMD EPYC Genoa

NVMe PCIe 5.0

3x Replication

10Gbps Network

VEEAM EXCLUSIVE

veeam

Hybrid Cloud Backup E-Book

Mastering cost control
and security

LEARN MORE



AUTOMATION PANEL EQUIPMENT

EXPLAINED

1 Power Filter

Filters electrical noise and improves power quality.

5 Solid State Relay

Switches loads electronically without moving parts.

11 Fuse

Protects circuits from overcurrent and short circuit.

16 CPU

The brain of the automation system.

21 Electrical Rail

DIN rail for mounting and organizing components.

24 Fan

Maintains proper airflow and temperature inside the panel.

2 Surge Protector

3 Polarized Circuit Breaker

4 Air Switch

Manually isolates power to the circuit for safety and maintenance.

10 Rail Socket

Provides power supply for maintenance tools and equipment.

5 Solid State Relay

6 Contactor

7 Motor Protector

8 Thermostat

9 Rail Socket

11 Fuse

12 Relay

13 Switched Power Supply

14 DC-DC Converter

15 Solid State Relay

17 I/O Expansion Module

18 Pulse Relay

19 Safety Relay

21 Terminal Block

22 Wiring Pin

20 Switch

Provides network connectivity between devices.

27 Remote I/O

Interfaces field signals from remote locations.

24 Fan

25 Inverter

26 Small Inverter

27 Remote I/O

WHY THESE COMPONENTS ARE IMPORTANT?



- ✓ Ensure safety, protection, and reliable operation
- ✓ Improve system performance and accuracy
- ✓ Reduce downtime and maintenance
- ✓ Provide clean, stable power and communication
- ✓ Make troubleshooting easier

TIPS FOR PANEL MAINTENANCE

- ✓ Keep panel clean and dust-free
- ✓ Check all connections regularly
- ✓ Ensure proper ventilation and cooling
- ✓ Verify tightness of terminals
- ✓ Follow wiring diagrams and safety procedures



REMEMBER: A well-designed panel ensures a reliable and efficient automation system.

Atur Perusahaan dalam 1 Aplikasi, Emang Bisa?

Masih pakai banyak software/aplikasi buat ngurus perusahaan? Kolabo punya semua ga pakai ribet!

HRM

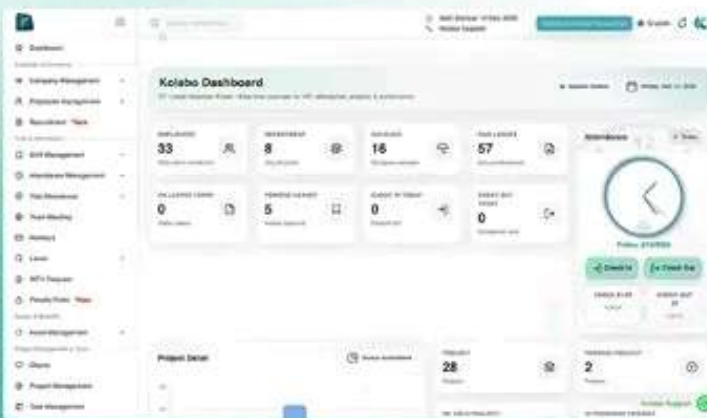
Accounting

CRM

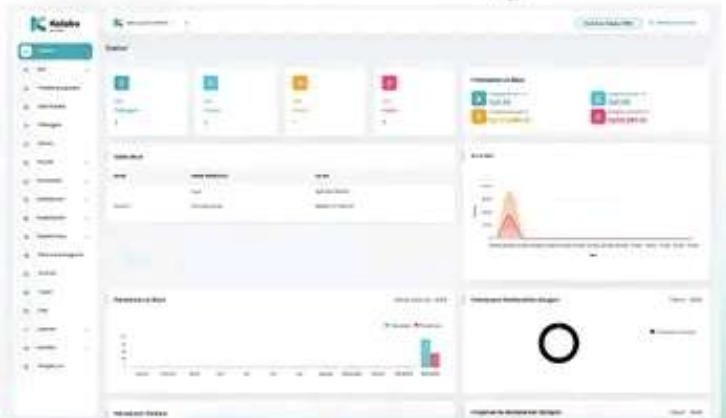
Project & Task Management



Dashboard HRM



Dashboard Accounting



400+ Sinyal Perangkat. Tanpa Hambatan. Kontrol Total.

Inteligensi perangkat waktu-nyata yang melindungi bisnis Anda tanpa memperlambat pelanggan.





SENTINEL X

SUDAH KENAL SENTINEL-X?

Saatnya Cari Tahu Cara Kerjanya

CEK INFORMASI >





HR itu **Human Resources** **Bukan** Human Robot

HR bukan robot yang kerjanya cuma input data. Biarkan Gadjian ambil tugas admin anda, semua urusan beres dalam sekali klik.



**Hitung
Gaji**



**Hitung PPH 21
& BPJS**



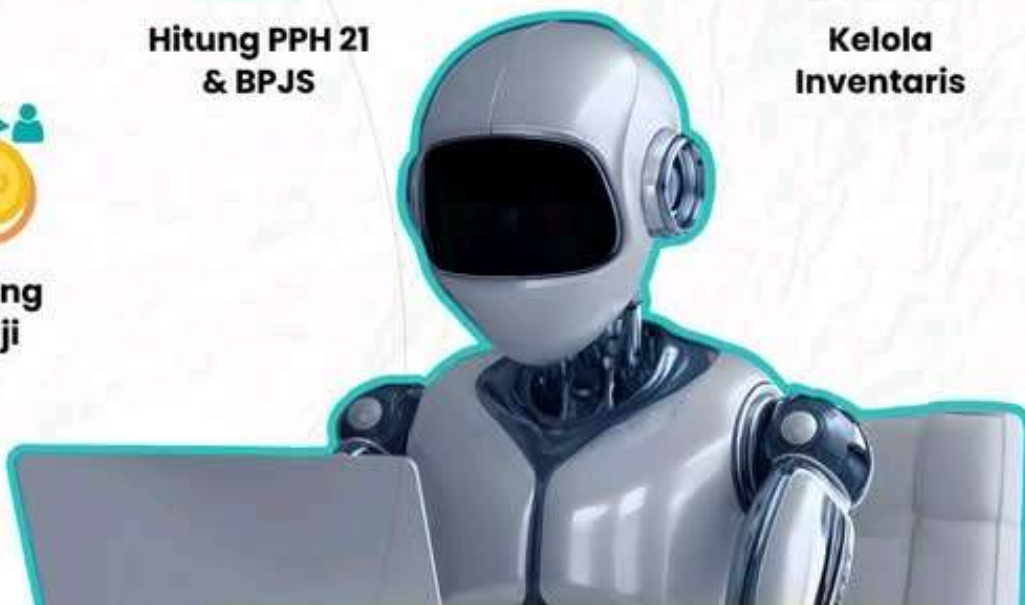
**Kelola
Absensi**



**Kelola
Inventaris**



**Kelola
Rekrutmen**



VIRTUAL HANDS ON LAB

AGENTIC AI IMMERSION WORKSHOP

30 July

9:30 AM IST | 12:00 PM SGT

2:00 PM AEST

REGISTER NOW



SHEENA NASIM

Principal Architect, Machine Learning
Field CTO APJ
Snowflake

LIVE &
LOCAL



The power of AI and ML in security.

See how leading companies
use it.

[Read e-book](#)



Agentic AI is the future...

and the future of agentic AI is
event-driven

[Download eBook](#)



A

Apex Wholesale

WORKSPACE

Dashboard

Apex Wholesale

WORKSPACE

Dashboard

Contacts

Companies

Deals

Analytics

Calendar

Inbox

Settings

Dave Anderson

HomeDashboard

Search

ReportsSettingsIntegrationsCompeles

HomeDashboard

Search

Dashboard

REVENUE

\$541k12.5%

ACTIVE DEALS

288%

WIN RATE

34.2%2.1%

AVG DEAL

\$89k3.2%

Revenue

6M1YAll

\$180k

\$135k

\$90k

Feb

Pipeline

Discovery

Proposal

Negotiation

Closed

\$0k

\$95k

\$100k

\$320k

Activity

View all

James Wilson

Follow-up on bulk order

2 hours ago

Emily Chen

Sent pricing sheet

4 hours ago

Michael Brooks

Contract review meeting

Yesterday

Recent Deals

View all

Q1 Bulk Order

Martin's Hardware

\$125k

Negotiation

Feb 2

Spring Inventory

City Builders Supply

\$89k

Proposal

Jan 31

Annual Contract

Midwest Retail Group

\$340k

Closed

Jan 29

NO API KEYS.

NO DEVOPS.

NO AGENCIES.

NO KIDDING.

Hercules



SMOKE EXHAUST FAN SIZING

STEP-BY-STEP CALCULATION (AS PER NBC & NFPA)



Objective: To remove smoke and hot gases during fire and maintain tenable conditions for safe evacuation.



1. DESIGN BASIS

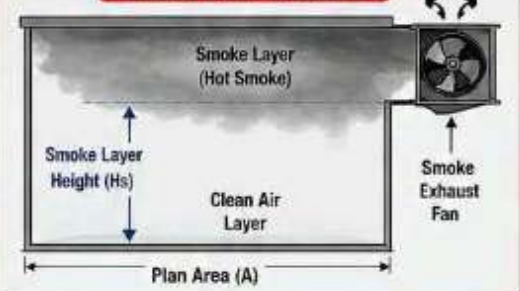
- ✓ Provide smoke exhaust during fire
- ✓ Maintain layer of clean air for evacuation
- ✓ As per NBC 2016, Part 4 & NFPA 92
- ✓ High temperature rated (300°C / 2 Hours)
- ✓ Power supply through fire rated circuit



2. INPUT DATA (EXAMPLE)

- Area to be protected (A) : 500 m²
- Ceiling Height (H) : 4.0 m
- Smoke Layer Height : 2.5 m
- Volume of Smoke Zone (V) : $A \times H_s$
= $500 \times 2.5 = 1,250$ m³
- Air Changes per Hour (ACH) : 10 ACH
- Discharge Coefficient (C_d) : 0.65
- Opening Type : Natural Opening (Louvers)
- Number of Fans : 2 Nos.

SMOKE ZONE CONCEPT



3. AIRFLOW (Q) CALCULATION

Method 1: By Air Changes per Hour (ACH)

$$Q = \frac{ACH \times V}{60} \quad (\text{m}^3/\text{s})$$

Where,
 ACH = Air Changes per Hour
 V = Volume of Smoke Zone (m³)
 Q = Airflow (m³/s)

$$Q = \frac{10 \times 1,250}{60} = 208.3 \text{ m}^3/\text{s}$$

Adopt Higher Value of Q
Q = 210 m³/s (Say)

Method 2: By Plume Flow (NBC Method)

$$Q = C_d \times A_o \times \sqrt{2 g H_s (T_s - T_a) / T_s} \quad (\text{m}^3/\text{s})$$

Where,
 C_d = Discharge Coefficient (0.60 – 0.65)
 A_o = Effective Opening Area (m²)
 H_s = Smoke Layer Height from Opening (m)
 T_s = Smoke Temperature (K) ≈ 473 K (200°C)
 T_a = Ambient Temperature (K) ≈ 293 K (20°C)
 g = Acceleration due to Gravity = 9.81 m/s²

Example Calculation:

C_d = 0.65
 A_o = 2.0 m²
 H_s = 2.5 m
 T_s = 473 K
 T_a = 293 K

$$Q = 0.65 \times 2.0 \times \sqrt{\frac{2 \times 9.81 \times 2.5 \times (473 - 293)}{473}}$$

$$Q = 210.2 \text{ m}^3/\text{s}$$

4. FAN CAPACITY (TOTAL)

Total Airflow Required = 210 m³/s

Number of Fans = 2 Nos.

$$Q_{\text{fan}} = \frac{Q_{\text{total}}}{N}$$

$$Q_{\text{fan}} = \frac{210}{2} = 105 \text{ m}^3/\text{s}$$

Select each fan for 105 m³/s capacity



5. STATIC PRESSURE (ESP) ESTIMATION

Consider losses through:

- Duct friction loss
- Louvers loss
- Elbows / bends
- Outlet loss
- Safety factor



Typical ESP for Smoke Exhaust System
 = 75 to 150 Pa
 (0.75 to 1.5 in. wg)

LOSS COMPONENT	TYPICAL LOSS (Pa)
Duct Friction Loss	40 – 60
Louvers Loss	20 – 30
Elbows / Bends Loss	10 – 20
Outlet Loss	10 – 20
Safety Factor (20%)	20 – 30
TOTAL ESP (ADOPT)	100 – 130 Pa

6. FAN SELECTION

- Airflow per fan : 105 m³/s
- External Static Pressure : 100 Pa (Adopt)
- Temperature Rating : 300°C / 2 Hours
- Type : Centrifugal / Axial Smoke Exhaust Fan
- Drive : Direct / Belt Driven
- Installation : Roof / Wall Mounted
- Select fan from approved manufacturer with certified performance curve.



7. POWER CALCULATION (APPROX.)

$$P = \frac{Q \times ESP}{\eta} \quad (\text{Watt})$$

Where,
 Q = Airflow (m³/s)
 ESP = Static Pressure (Pa)
 η = Fan Efficiency (Assume 0.6)

For each fan:

$$P = \frac{105 \times 100}{0.6}$$

$$P = 17,500 \text{ W}$$

$$= 17.5 \text{ kW (Approx.)}$$

8. SUMMARY

- ✓ Required Airflow (Total) : 210 m³/s
- ✓ Airflow per Fan : 105 m³/s
- ✓ ESP (Adopt) : 100 Pa
- ✓ Fan Type : HT Smoke Exhaust Fan
- ✓ Temperature Rating : 300°C / 2 Hours
- ✓ Number of Fans : 2 Nos.



IMPORTANT NOTES



All smoke exhaust fans must be 300°C / 2 Hours rated.



Use fire rated cables & power supply.



Provide motorized smoke dampers with fusible link.



Test and commission the system.



Follow NBC, NFPA 92 & Local Authority Requirements.

✓ **PROPER SMOKE EXHAUST SYSTEM SAVES LIVES AND ENSURES SAFE EVACUATION.**



DANGER
HIGH VOLTAGE

33kV to 11kV

DISTRIBUTION TRANSFORMER SUBSTATION LAYOUT



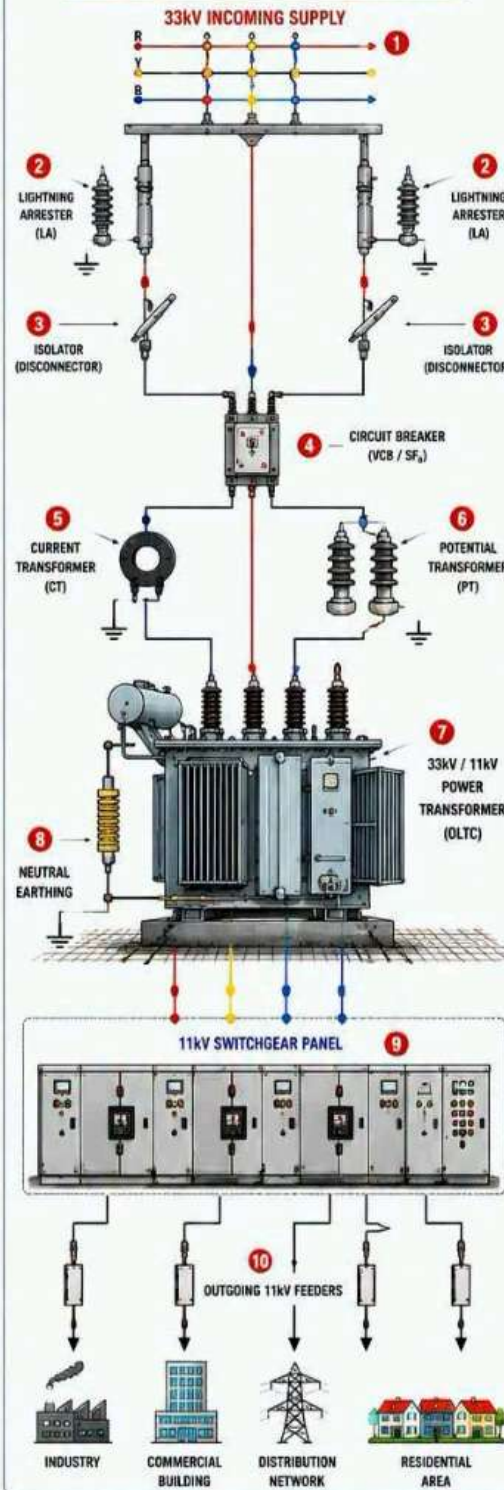
SAFETY FIRST
POWER ALWAYS

COMPONENTS & FUNCTIONS

- 33KV INCOMING SUPPLY**
Power is received from the utility through 33KV overhead line.
- LIGHTNING ARRESTER (LA)**
Protects equipment from lightning surges and overvoltages.
- ISOLATOR (DISCONNECTOR)**
Provides visible isolation for maintenance and safety.
- CIRCUIT BREAKER (VCB/SF₆)**
Interrupts fault current and protects the equipment.
- CURRENT TRANSFORMER (CT)**
Steps down high current to 1A or 5A for protection and metering.
- POTENTIAL TRANSFORMER (PT)**
Steps down high voltage to 110V for metering and protection.
- POWER TRANSFORMER**
33KV/11KV transformer steps down voltage with OLTC (On Load Tap Changer) to maintain voltage.
- NEUTRAL EARTHING**
Transformer neutral is earthed through earthing resistor or solid earthing.
- 11KV SWITCHGEAR**
11KV panels receive power from transformer and feed to outgoing circuits.
- OUTGOING 11KV FEEDERS**
Power is distributed to industries, commercial loads and distribution network.
- CONTROL & PROTECTION PANEL**
Houses relays, meters, SCADA, annunciators and control devices.
- STATION AUXILIARY SUPPLY**
Provides AC/DC supply for control, protection, lighting and other auxiliary loads.
- EARTHING SYSTEM**
Equipotential earthing ensures safety of equipment and personnel.

STEP DOWN • PROTECT • CONTROL • DISTRIBUTE

SUBSTATION SINGLE LINE DIAGRAM (33kV to 11kV)



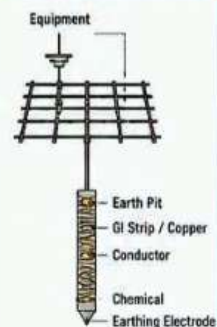
POWER FLOW SEQUENCE

- 33kV Supply from Utility
- Lightning Arresters Protect from Surges
- Isolators Provide Isolation
- Circuit Breaker Protects from Faults
- CT & PT for Metering & Protection
- Step Down by 33/11KV Transformer
- Power at 11kV Level
- 11kV Switchgear Distribution
- Power to Outgoing Feeders

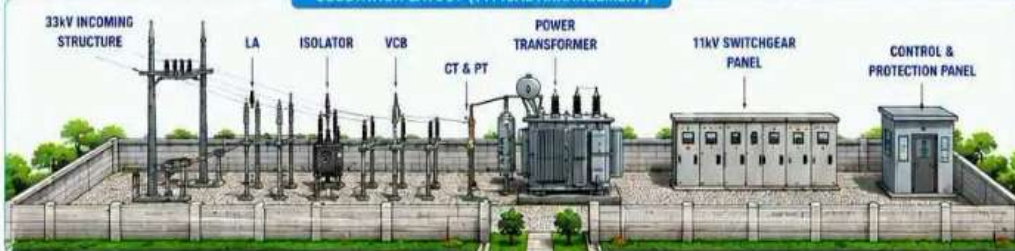
KEY SPECIFICATIONS

- Nominal Voltage (HV) : 33KV
- Nominal Voltage (LV) : 11KV
- Transformer Rating : As per requirement (Example ~ 5MVA / 10MVA)
- Frequency : 50 Hz
- Cooling : ONAN / ONAF
- Vector Group : Dyn11
- Impedance : 6% (Typical)
- Earthing System : Solid / Resistive Earthing

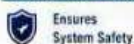
EARTHING SYSTEM DIAGRAM



SUBSTATION LAYOUT (TYPICAL ARRANGEMENT)



IMPORTANCE OF SUBSTATION



Ensures System Safety



Provides Reliable Power



Facilitates Easy Maintenance



Improves Power Quality



Supports Industrial & Commercial Growth